CSSEMA 384 – Data Mining with Programming
Rose-Hulman Institute of Technology

Worksheet 02: Data Acquisition and Cleaning

Name (Print):_____ Section:_____

Topic: Data Acquisition and Cleaning

Work with a neighbor. Circle or select answers as directed. Fill only the short blanks; no coding is required.

1. **Collecting data.** Match each method to the example. Write A, B, or C; use each once.

   A. API B. Web scraping C. Logging

   (a) A campus app records each time its own bike rack sensor reports a count. _____

   (b) A documented service returns bike availability when queried. _____

   (c) A program extracts bike counts from an HTML table. _____

   Circle the stronger source for comparing *this term's* campus bike use:

   this term's campus sensor records / last year's national car sales.

2. **Error or processing artifact?** Circle one per row.

What happened?	**Circle one**
A failed sensor never recorded two hours of readings.	Acquisition error / Artifact
An import script multiplied every count by 10; raw counts remain intact.	Acquisition error / Artifact

   Circle all good practices: keep raw data / clean a copy / overwrite raw data / record cleaning steps.

3. **Make data compatible.** Circle the best action on each line.

   (a) One file uses meters; another uses centimeters:
   convert to a common unit / combine the numbers unchanged.

   (b) Two people share the name Alex Lee:
   merge by name alone / check stable IDs before merging.

   (c) Two logs use different time zones:
   align to a common time basis / compare clock labels unchanged.

4. **Missing is not zero.** A bike-count file defines `NA` and `-999` as missing-value codes. A recorded count of `0` means the rack was empty.

   Circle all missing entries: `12` `NA` `0` `-999`

   Replace a missing count with zero automatically? Yes / No.

   Before averaging: recognize missing codes / include every code.

5. **Delete or investigate?** Sensors fail more often when racks are crowded. Circle the best answer.

   (a) Dropping all missing records could make estimated use:
   too low / guaranteed correct / unaffected in every case.
   (b) Only 5% of records are missing. Is deletion automatically safe? Yes / No.
   (c) Before choosing a treatment, check:
   why values are missing / only the file size.

6. **Imputation.** Recorded counts are 4, 8, and 12; one more count is missing. For this exercise, use the observed mean to fill it.

   Circle the imputed value: 0 / 8 / 12.

   After filling it, the four-value mean is: 6 / 8 / 10.

   Circle one: The filled value is an *observation / estimate*.

   Does keeping the mean guarantee no other distortion? Yes / No.

7. **Z-scores and scale.** Use $z = (x - \mu)/\sigma$. A reference dataset has mean $\mu = 10$ and standard deviation $\sigma = 2$.

   (a) For $x = 14$, $z =$ -2 / 0 / 2.
   (b) For $x = 8$, $z =$ -1 / 1 / 4.
   (c) A negative z-score means: below the mean / a data error.
   (d) Are all z-scores between 0 and 1? Yes / No.
   (e) Does standardizing automatically make any distribution bell-shaped? Yes / No.

8. **Do units change a z-score?** Height: $x = 180$ cm, $\mu = 170$ cm, $\sigma = 10$ cm.

   Circle its z-score: 1 / 10 / 180.

   In meters: $x = 1.80$, $\mu = 1.70$, $\sigma = 0.10$. Circle its z-score: 0.01 / 1 / 100.

   Z-scores have: the original measurement units / no measurement units.

9. **Compare different scales.** Use the reference statistics below.

Feature	**Value**	**Mean**	**Std. dev.**
Bike count	120	100	20
Wait time (minutes)	14	10	2

   Farther above its own mean, in standard deviations: bike count / wait time / equally far.

   After standardizing each reference dataset, circle its mean: *0 / 1*; standard deviation: *0 / 1*.

10. **Reflection.** Circle one interesting topic and underline one to revisit:

    data collection artifacts compatibility missing values z-scores

    Circle one helpful next step: another example / class discussion / more practice.
    Any questions? Write below, or circle *None*.
